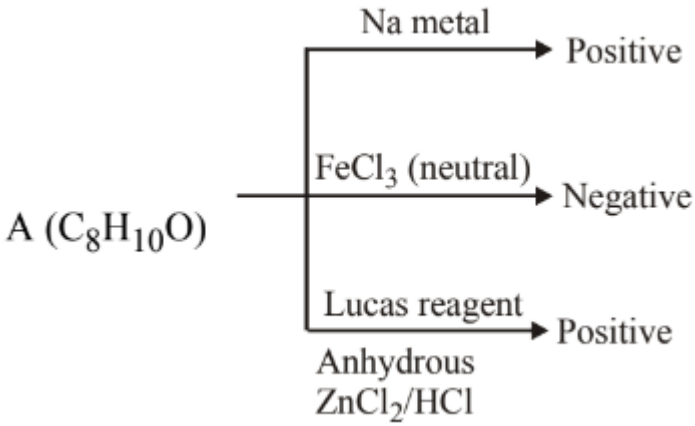
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

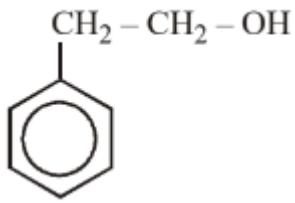
Q.1

An aromatic compound $A\ (C_8H_{10}O)$ gives following tests with the given reagents.

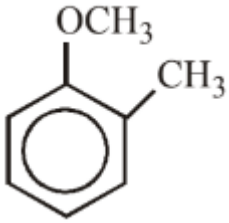


Identify A -

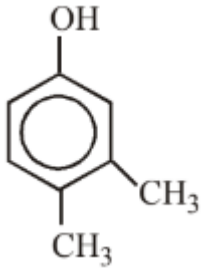
(A)



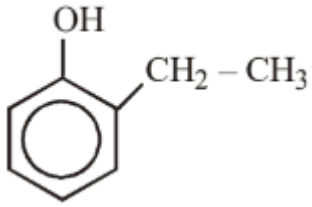
(B)



(C)



(D)



Q.2

Which of the following reagent is used for the separation of acetaldehyde from acetophenone?

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(A) NH_2OH

(B) $NaOI$

(C) Tollens reagent

(D) $C_6H_5NHNH_2$

Q.3

In Kjeldahl's method, nitrogen present is estimated as

(A) N_2

(B) NH_3

(C) NO_2

(D) None of these.

Q.4

Distillation is used to separate liquids which differ in the b.p. by

(A) $5^\circ C$

(B) $10^\circ C$

(C) $30 - 80^\circ C$

(D) $100^\circ C$

Q.5

In Duma's method, nitrogen present in organic compound is estimated as

(A) N_2

(B) NO

(C) NH_3

(D) H_2

Q.6

Aniline is purified by

(A) Steam distillation

(B) Simple distillation

(C) Vacuum distillation

(D) Extraction with a solvent

Q.7

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An organic compound contains about 52% carbon. It could be

(A) only ethanol

(B) only dimethyl ether

(C) both

(D) none of these

Q.8

Soda lime test is used to detect the following element of organic compound

(A) C

(B) H

(C) N

(D) S

Q.9

Phthalic anhydride can be purified by-

(A) Recrystallisation

(B) Distillation

(C) Sublimation

(D) Acidification and regeneration

Q.10

A mixture of acetone and methanol can be separated by -

(A) Vaccum distillation

(B) Steam distillation

(C) Fractional distillation

(D) None of these

Q.11

An organic compound has been found to possess the Empirical formula $\mathrm{CH_2O}$ and molecular weight 90 . The molecular formula of it is $(\mathrm{C}=12, \mathrm{H}=1$ and $\mathrm{O}=16)$

(A) $\mathrm{C_3H_6O_3}$

(B) $\mathrm{CH_2O}$

(C) $\mathrm{C_3H_6O_2}$

(D) $\mathrm{C_2H_2O}$

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Q.12

An organic compound containing carbon hydrogen and oxygen contains 52.20% carbon and 13.04% hydrogen. Vapour density of the compound is 23 . Its molecular formula will be

(A) C_2H_6O

(B) C_3H_8O

(C) C_4H_8O

(D) $C_5H_{10}O$

Q.13

Quantitative measurement of nitrogen in an organic compounds is done by the method

(A) Berthelot method

(B) Belstein method

(C) Lassaigne test

(D) Kjeldahl's method

Q.14

The best method for the separation of naphthalene and benzoic acid from their mixture is

(A) Chromatography

(B) Crystallisation

(C) Distillation

(D) Sublimation

Q.15

Which of the following pair of the species has the same percentage of carbon

(A) CH_3COOH and C_2H_5OH

(B) $C_6H_{12}O_6$ and $C_{12}H_{22}O_{11}$

(C) $HCOOCH_3$ and $C_{12}H_{22}O_{11}$

(D) CH_3COOH and $C_6H_{12}O_6$

Q.16

A mixture of camphor and benzoic acid can be separated by

(A) Chemical method

(B) Sublimation

(C) Fractional distillation

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(D) Extraction with a solvent

Q.17

X (molecular formula, $C_7H_6O_2$) is an aromatic white solid which liberates colourless, odourless, gas on reacting with $NaHCO_3$ then choose the correct options -

(A) Only three of the five functional isomers of X (including X itself) will give positive 2,4-DNP test

(B) The liberated colourless, odourless gas will containing radioactive ^{14}C.

(C) Except X, no other functional isomer will liberate colourless odourless gas with $NaHCO_3$

(D) The DU of higher homolog of X will be four.

Q.18

Phenol cannot be distinguished from ethanol by

(A) Tollen's reagent

(B) Benedict's reagent

(C) $FeCl_3$

(D) Schiff's base

Q.19

Sodium extract of an organic compound gives blood red colour with $FeCl_3$. It does not contain $-$

(A) N

(B) S

(C) Cl

(D) Br

Q.20

Passage:

Observe the following sequence of reactions:

$$P\,(C_9H_9Br) \xrightarrow[\text{reductive}]{O_3/Zn,H_2O} R + Q \xrightarrow[(ii)\ H^+]{(i)\ \text{Tollen's reagent}} (s) + Ag\downarrow$$

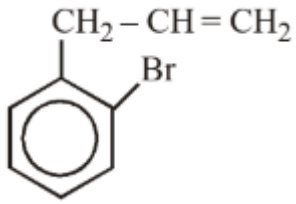
P shows geometrical isomerism. Q gives positive Tollen's test and the oxidation product of Tollen's test followed by acidification is the strongest acid among its all position isomers. R gives positive lab tests with 2,4-DNP, Fehling solution and $I_2/NaOH$ reagents.

Question:

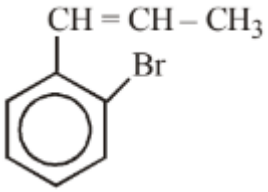
The compound P can be $-$

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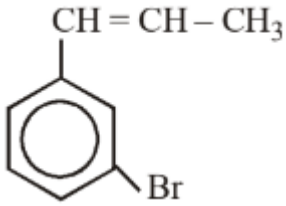
(A)



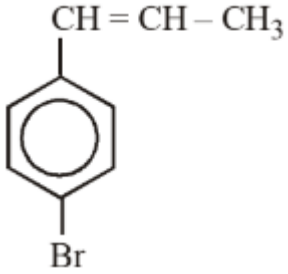
(B)



(C)



(D)

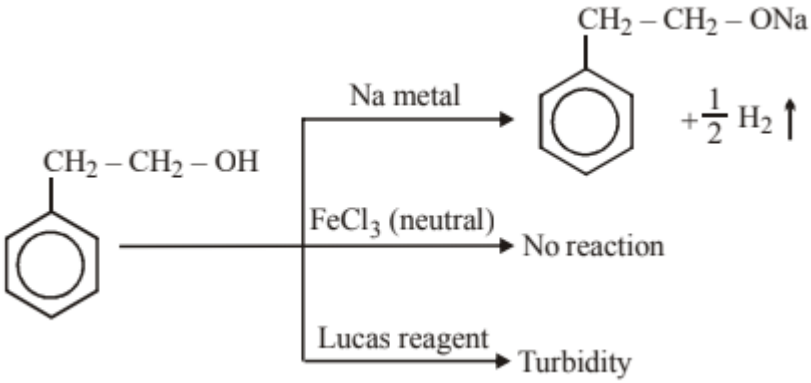


Answers & Solutions

Q.1 Answer:

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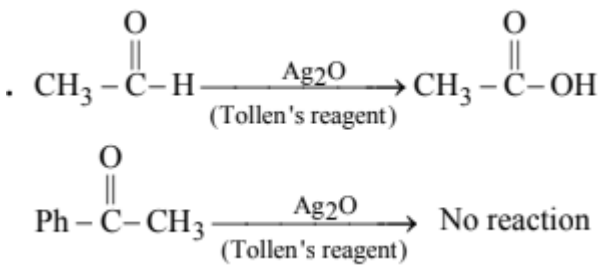
Solution:



Q.2 Answer:

Tollens reagent

Solution:



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Q.3 Answer:

NH_3

Solution:

In Kjeldahl's method, nitrogen present is estimated as NH_3.

Q.4 Answer:

$30-80^{\circ}C$

Solution:

Distillation is used to separate liquids which differ in the b.p. by $30-80^{\circ}C$.

Q.5 Answer:

N_2

Solution:

Dumas' method involves the determination of nitrogen content in the organic compound in the form of nitrogen (N_2). Organic compound is oxidized with cupric oxide in an atmosphere of carbon dioxide which gives free nitrogen along-with carbon dioxide and water.

Q.6 Answer:

Steam distillation

Solution:

Aniline is an organic compound having chemical formula of $C_6H_5NH_2$. It is steam volatile and it is insoluble in water. Aniline is the simplest aromatic compound.

Q.7 Answer:

both

Solution:

C_2H_5OH

$\%C=\frac{24}{46}\times 100=52.17\%$

CH_3-O-CH_3

$\%C=\frac{24}{46}\times 100=52\%$

Q.8 Answer:

N

Solution:

Soda lime test: A small amount of an organic compound is heated strongly with soda lime $(CaO+NaOH)$. The liberation of ammonia indicates the presence of nitrogen in the organic compound.

$$\underset{\text{Acetamide}}{CH_3CONH_2}+NaOH\xrightarrow{CaO}\underset{\text{Sodium acetate}}{CH_3COONa}+NH_3\uparrow$$

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Q.9 Answer:

Sublimation

Solution:

This invention relates to a method of purifying phthalic anhydride, particularly a crude or a partially purified phthalic anhydride which has been produced by the catalytic oxidation of napthalene or other organic substance in the vapor phase. In the production of phthalic anhydride by .gsubjecting napthalene in the vapor phase to partial or selective oxidation by means of air or other oxygen-containing gas in the presence of a suitable catalyzer at an appropriate temperature, it .is well'known that the phthalic anhydride pro qduced is usually contaminated with various impurities. The bulk of these impurities can usu- ,3 allybe removed by distillation or by holding the 3;;cr1'1de phthalic anhydride at or near the boiling point for some time, followed by distillation.

Q.10 Answer:

Fractional distillation

Solution:

The mixture of liquids can be separated by the Distillation process. The process involves the heating of the mixture so that the component with low boiling point escapes first followed by the other. The acetone and methanol are volatile liquids. The boiling point of liquids is close to each other. Thus, a suitable and sophisticated distillation technique is used to separate the liquids.

Q.11 Answer:

$C_3H_6O_3$

Solution:

Empirical formula $= CH_2O$

Empirical weight $= 12 + 2 + 16 = 30$

Molecular weight $= 90$

$n = \frac{\text{Molecular weight}}{\text{Empirical weight}} = \frac{90}{30} = 3$

Molecular formula $= C_3H_6O_3$

Q.12 Answer:

C_2H_6O

Solution:

Percentage of O **is** 34.76.

$\therefore C : H : O = \frac{52.20}{12} : \frac{13.04}{1} : \frac{34.76}{16} = 4.33 : 13 : 2.15 = 2 : 6 : 1$

Equivalent weight $= 46$

Molecular weight $= 2 \times 23 = 46.$

$x = \frac{46}{46} = 1$

Molecular formula $= (\text{ Empirical formula })_{\mathbf{x}}$

Molecular formula $= C_2H_6O$

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Q.13 Answer:

Kjeldahl's method

Solution:

Kjeldahl's and Duma's methods are used for the quantitative estimation of nitrogen in an organic compound. In the Kjeldahl method, the nitrogen element of organic compound is changed to the ammonia.

Q.14 Answer:

Sublimation

Solution:

Mixture of benzoic acid and naphthalene can be separated from hot water in which benzoic acid dissolves but naphthalene does not.

Q.15 Answer:

CH_3COOH and $C_6H_{12}O_6$

Solution:

Molecular mass of $CH_3COOH = 60$

$$\therefore \% \text{ of C in } CH_3COOH = \frac{24}{60} \times 100 = 40\%$$

Similarly,

Molecular weight of $C_6H_{12}O_6 = 180$

$$\therefore \% \text{ of C in } C_6H_{12}O_6 = \frac{72}{180} \times 100 = 40\%$$

Q.16 Answer:

Chemical method

Solution:

- Benzoic acid is an acid and therefore reacts with a base but camphor is an organic compound that does not react with the base.

- Aqueous solution of sodium bicarbonate $(NaHCO_3)$ **is used to separate the mixture of benzoic acid and camphor.**

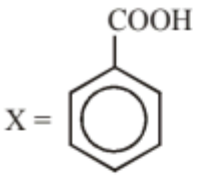
- Benzoic acid reacts with an aqueous solution of sodium bicarbonate to form sodium benzoate (C_6H_5COONa) **which is water-soluble whereas camphor does not react with sodium bicarbonate.**

Q.17 Answer:

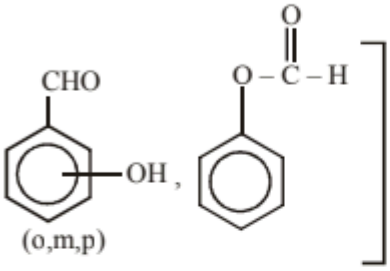
The liberated colourless, odourless gas will containing radioactive ^{14}C.

Solution:

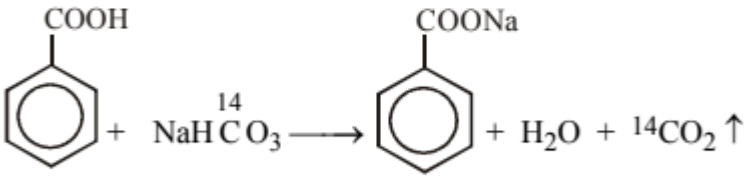
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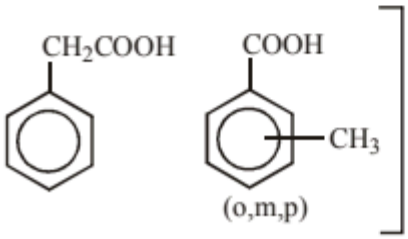
Other functional isomers of X are



None of these will give +ve test with $NaH^{14}CO_3$



Higher homolog of X :



4 isomeric form

Q.18 Answer:

Tollen's reagent

Solution:

Phenol can be distinguished from ethanol by $FeCl_3$.

Q.19 Answer:

Cl

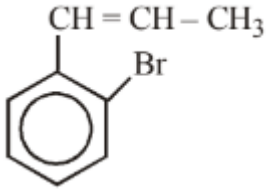
Solution:

$$Na + N + C + S \rightarrow \underset{\text{Sodium thiocyanate}}{NaSCN}$$

$$SCN^- + Fe^{3+} \rightarrow \underset{\text{Ferric thiocyanate (Blood-red colour)}}{[Fe(SCN)]^{2+}}$$

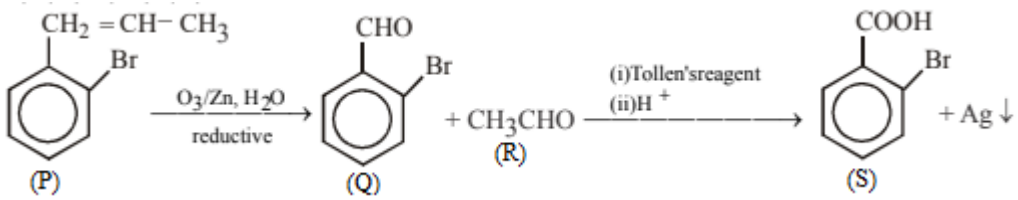
It confirms the presence of both N and S.

Q.20 Answer:



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Solution:



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